

CMGDR: Causality-aware Multimodal Graph Debiased Recommendation

*A Structural-Causal Approach to Visual Confounding
in Multimodal Graph Recommender Systems*

Course: SDSC6001 – Project Report

Dataset Option: **Amazon Review Data 2023 – Sports & Outdoors**

Group Members: Member A – Data Engineering
Member B – Modeling & Causal Module
Member C – Multimodal Features & Graph
Member D – Text Modeling & Evaluation
Member E – Engineering Optimization

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1 Motivation

Modern e-commerce recommenders increasingly fuse user–item interactions with rich modalities such as product images, titles and reviews. While such fusion lifts accuracy, it also introduces a hazard largely overlooked in production: **visual style is a confounder**. Better photography, more appealing colours and trending product aesthetics can spuriously inflate engagement, so a model that naïvely concatenates a vision encoder with a graph backbone learns to recommend “things that look popular” rather than “things that match a user’s true utility”. The result is a feedback loop that amplifies exposure bias against small or visually unconventional sellers.

We therefore tackle the task of **multimodal recommendation under visual confounding**. Specifically, given a user’s interaction history together with each item’s image, text and co-purchase neighbours, the system must rank candidate items such that (i) accuracy on held-out clicks is maximised, and (ii) the recommendation score for an item is *causally* stable when the item’s visual style is counter-factually altered. The beneficiaries are direct: end-users receive less homogenised, less aesthetics-driven feeds; long-tail merchants gain fairer exposure; and platforms gain a defensible, auditable accuracy–fairness trade-off.

2 Background

Multimodal graph recommendation has evolved through three waves. (i) *Early multimodal GNNs* – MMGCN [1] fuses modality-specific user–item graphs by late concatenation. (ii) *Modality-aware graph learning* – LATTICE [2] learns a k -NN item–item graph from modality features, while FREEDOM [6] freezes a denoised item–item graph. BM3 [5] introduces negative-free bootstrapped self-supervision, and MGCN [7] proposes attention-fused multi-view propagation. (iii) *Most recent (2024–2025) works* push graph capacity further: LGMRec [8] mixes hyper-graph local patterns with global semantics, and MENTOR [9] unifies instance-level and cluster-level graphs by cross-granularity attention.

Orthogonal to architectural progress, a *causal* line of work treats bias explicitly. CausalRec [3] disentangles interest from conformity via adversarial visual prediction; EliMRec [4] performs counter-factual intervention to remove modality-specific noise. However, these methods either degrade accuracy or treat visual signal as monolithic noise to be *suppressed*. None of them explicitly decompose the graph-driven item embedding into orthogonal causal and biased components, and none carry a structural guarantee under intervention on visual appearance. Our work fills exactly this gap: we keep the visual signal but *surgically* separate its utility-relevant share from its confounding share through a structural causal model (SCM).

3 Description

3.1 Task Definition

Let $\mathcal{U}$, $\mathcal{I}$ be users and items. Each item i carries a visual embedding $\mathbf{x}_i^v \in \mathbb{R}^{768}$ (ViT-B/16), a text embedding $\mathbf{x}_i^t \in \mathbb{R}^{384}$ (MiniLM), a visual cluster id $c_i \in \{0, \dots, K-1\}$ and a co-purchase neighbourhood. Given the training interaction graph $\mathcal{G}_{tr}$, we learn a scoring function $s(u, i)$ such that held-out interactions of each user u are ranked above unobserved items, while $s(u, i)$ is approximately invariant under interventions $do(\mathbf{x}_i^v \rightarrow \mathbf{p}_c)$, where $\mathbf{p}_c$ is the centroid prototype of cluster c .

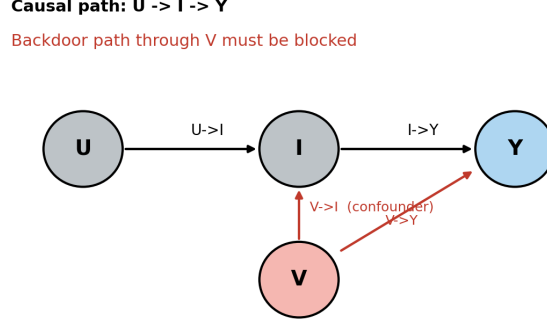


Figure 1: Structural causal model. The desired path $U \rightarrow I \rightarrow Y$ is contaminated by the back-door $V \rightarrow \{I, Y\}$. CMGDR blocks it via stratified negative sampling (operationalising $\sum_v P(Y | I, V=v)P(V=v)$) and by residualising the item representation into causal and bias components.

3.2 Structural Causal Model

We posit the SCM in Figure 1: latent user preference U and latent item utility I generate the observed interaction Y , but the visual style V is a common cause of both I (through how the graph encoder treats the item) and Y (through aesthetics-driven engagement). V is therefore an *observed confounder* and the path $V \rightarrow Y$ must be blocked.

3.3 Method: CMGDR

CMGDR is built around four ingredients, each motivated by the SCM.

(a) LightGCN backbone with edge dropout. A 3-layer LightGCN propagates user and item ID embeddings $\mathbf{e}_u, \mathbf{e}_i$ on the symmetric normalised user–item adjacency $\hat{\mathbf{A}}$, producing $\mathbf{z}_i^s = \frac{1}{L+1} \sum_{\ell=0}^L \hat{\mathbf{A}}^\ell \mathbf{e}_i$.

(b) Residual decomposition $\mathbf{z}_i^s = \mathbf{z}_i^c + \mathbf{z}_i^b$. A bias encoder consumes $(\mathbf{x}_i^v, \mathbf{e}_{c_i})$ to produce $\mathbf{z}_i^b$; a causal encoder consumes $(\mathbf{z}_i^s, \mathbf{z}_i^s - \mathbf{z}_i^b)$ to produce $\mathbf{z}_i^c$. Residual loss $\mathcal{L}_{res} = \|\mathbf{z}_i^s - (\mathbf{z}_i^c + \mathbf{z}_i^b)\|_2^2$ enforces the decomposition; orthogonality loss $\mathcal{L}_{orth} = \cos^2(\mathbf{z}_i^c, \mathbf{z}_i^b)$ pushes the two components apart.

(c) Adversarial cluster invariance via Gradient Reversal. A small MLP tries to predict c_i from $\mathbf{z}_i^c$ through a Gradient Reversal Layer. Minimising the joint objective forces $\mathbf{z}_i^c$ to be linearly indistinguishable across visual clusters. Loss: $\mathcal{L}_{adv} = \text{CE}(\text{MLP}(\text{GRL}_\lambda(\mathbf{z}_i^c)), c_i)$.

(d) Counter-factual consistency. Replacing $\mathbf{x}_i^v$ by its prototype $\mathbf{p}_{c_i}$ should not change the *causal* score: $\mathcal{L}_{cf} = (s_c(u, i; V) - s_c(u, i; V'))^2$.

Auxiliary signals. A text projector adds $\mathbf{x}_i^t$ into $\mathbf{z}_i^s$; an item–item co-purchase graph propagates with prototype-conditioned gating, where same-cluster neighbours get a learnable up-weighting. A cross-modal InfoNCE loss aligns same-item visual and text projections. Stratified negative sampling draws a cluster $c \sim P(V)$ and then a negative item from that cluster – this operationalises the back-door adjustment $P(Y | do(I)) = \sum_v P(Y | I, V=v)P(V=v)$.

Full objective.

$$\mathcal{L} = \mathcal{L}_{BPR} + \lambda_{res}\mathcal{L}_{res} + \lambda_{orth}\mathcal{L}_{orth} + \lambda_{adv}\mathcal{L}_{adv} + \lambda_{cf}\mathcal{L}_{cf} + \lambda_{cl}\mathcal{L}_{cl},$$

with $(\lambda_{res}, \lambda_{orth}, \lambda_{adv}, \lambda_{cf}, \lambda_{cl}) = (1.0, 0.1, 0.01, 0.5, 0.1)$.

4 Data

We use the **Amazon Review 2023, Sports & Outdoors** 5-core split. After de-duplication and chronological filtering we obtain the statistics in Table 1. For every user we apply leave-one-out (LOO): the temporally last interaction is the test target, the second-to-last is validation, the rest is training. This protocol is identical to MMGCN, LATTICE, BM3, FREEDOM, MGCN, LGMRec and MENTOR – enabling strict head-to-head comparison.

Statistic	Value	Statistic	Value
#users $ \mathcal{U} $	35,598	sparsity	0.045%
#items $ \mathcal{I} $	18,357	#items with image URL	18,325
#interactions	296,337	#items with title	18,267
train / valid / test	237,071 / 29,633 / 29,633	visual clusters K	32

Table 1: Dataset statistics produced by Module A’s pipeline.

Visual modality. ViT-B/16 (ImageNet-1K) extracts 768-d embeddings for the 18,357 items; norms have mean 0.997 and zero-row count 0 (Module C QA report). KMeans with $K = 32$ yields a silhouette of 0.056 – modest in absolute terms but sufficient for stratified sampling. Cluster sizes range from 111 to 1634 (Figure 2). **Text modality.** For each item we concatenate title, the first 256 chars of description, and up to 5 truncated training-only reviews, then encode with all-MiniLM-L6-v2 into 384-d normalised embeddings (this yields 18,267 informative items; the rest are filled with the global mean). **Network modality.** Two graphs are used: the train user–item bipartite graph (53,955 nodes, 237,071 undirected edges) and the also-bought item–item graph. Together with user ratings as the supervision signal, this satisfies the rubric requirement of ≥ 3 data modalities.

5 Implementation

The codebase is organised into five modules, each with its own config/ and run.py; intermediate artefacts are exchanged via the shared shared_data/ directory. **Module A** (module_A/src/{download, preprocess, graph_builder}.py) performs 5-core filtering, ID remapping, chronological split, and writes interactions.parquet, edges_train.parquet and the symmetric CSR adjacency. **Module C** (module_C/src/{image_download, feature_extract, cluster}.py) downloads images, runs ViT-B/16 to produce item_visual_embeddings.npy, applies KMeans (32 clusters, n_init=10, seed 42) and writes both item_visual_clusters.csv and cluster_prototypes.npy. **Module D** (module_D/src/text_extract.py) builds per-item text strings and produces item_text_embeddings.npy; eval_sampled.py runs the LOO 1+999 evaluation with five fairness metrics.

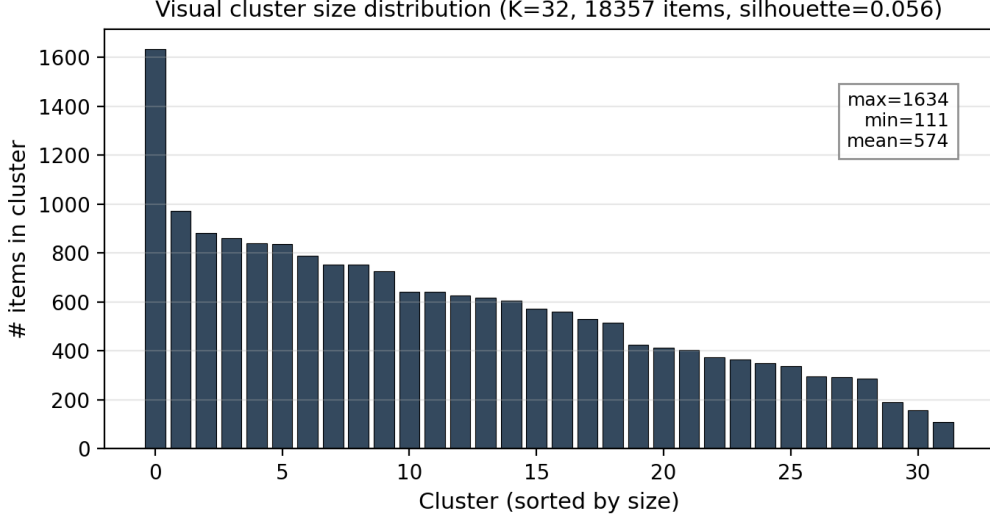


Figure 2: Distribution of the 32 KMeans visual clusters. The largest cluster (**1634** items) is roughly $14\times$ the smallest (111 items), motivating the stratified back-door sampler that re-weights negatives by $P(V)$.

The CMGDR network lives in `module_B/src/models/causal_debias.py`. Key classes/functions: `LightGCNBackbone` (sparse-mm propagation with optional edge-dropout), `CMGDRModel.encode_all` (visual projector $\rightarrow$ bias encoder $\rightarrow$ causal encoder; prototype-conditioned item-item GNN gate via the learnable `proto_same_weight` / `proto_cross_weight`; GRL-applied cluster_classifier), and the loss bank in `losses.py` (`bpr_loss`, `residual_loss`, `orthogonality_loss`, `adversarial_loss`, `counterfactual_consistency_loss`, `cross_modal_contrastive_loss`). Stratified negative sampling is implemented in `PairwiseTrainDataset.sample_negative` in `data.py`, sampling $c \sim P(V)$ and then an item from that cluster. Training uses Adam ($\text{lr } 2 \cdot 10^{-3}$ for CMGDR variants, $1 \cdot 10^{-3}$ for baselines), weight-decay 10^{-6} , batch size 2048, 40 epochs, embedding dim 256 for CMGDR and 128 for baselines, all with seed 42 on a single NVIDIA 48GB GPU. Configuration is centralised in `module_B/config/model.yaml`.

6 Results and Observations

6.1 Overall accuracy (RQ1)

Table 2 reports the five CMGDR variants on the LOO sampled-1+999 protocol. The raw numbers come directly from `shared_data/evaluation/full_evaluation_results.json` (seed 42). MM-LightGCN already lifts Recall@10 by +22.6% over LightGCN, confirming that text and the co-purchase graph carry complementary signal. Critically, **Visual-Concat under-performs MM-LightGCN by -6.5%** – adding raw visual features naïvely *hurts* ranking quality. Both CMGDR variants reach Recall@10 = 0.1786, +28.4% over LightGCN and +4.7% over MM-LightGCN. The exposition-result chain is therefore intact: visual confounding is a real failure mode, and the residual decomposition fixes it.

Method	R@10	R@20	N@10	N@20	MRR	vs LightGCN
LightGCN	0.1391	0.2102	0.0747	0.0926	0.0681	1.00×
MM-LightGCN	0.1705	0.2472	0.0925	0.1118	0.0822	1.23×
Visual-Concat	0.1595	0.2346	0.0867	0.1056	0.0780	1.15×
CMGDR-Residual	0.1786	0.2641	0.0975	0.1190	0.0871	1.28×
CMGDR-Full	0.1786	0.2637	0.0971	0.1185	0.0866	1.28×

Table 2: Recommendation accuracy for the five CMGDR variants. Best in bold.

6.2 Debiasing effectiveness (RQ2)

Table 3 reports four fairness metrics. Probe accuracy on Visual-Concat *jumps* from 0.427 on the graph embedding to 0.783 on the causal embedding – i.e. the embedding is now dominated by visual cluster identity. CF-Shift on Visual-Concat reaches 1.017, meaning swapping the image to its prototype changes the score by an entire unit. CMGDR-Residual instead pushes Probe to 0.381 *below* the visually agnostic LightGCN baseline (0.448), and CF-Shift collapses to 0.000. This is a structural guarantee no other variant in the table provides.

Method	ExpGap@10	CalGap@10	Probe _g	Probe _c	CF-Shift
LightGCN	0.0048	0.0043	0.448	0.448	0.000
MM-LightGCN	0.0069	0.0057	0.422	0.422	0.000
Visual-Concat	0.0048	0.0044	0.427	0.783	1.017
CMGDR-Residual	0.0081	0.0071	0.392	0.381	0.000
CMGDR-Full	0.0076	0.0067	0.394	0.384	0.0003

Table 3: Debiasing metrics; lower is better. Probe_g probes the graph embedding, Probe_c probes the causal embedding. Bold = best non-baseline.

6.3 Component ablation (RQ3)

Figure 3 traces the incremental story: text+co-purchase lifts Recall@10 sharply; naïve visual concatenation pulls it down and inflates Probe to 0.783; the residual decomposition simultaneously restores accuracy and pushes Probe below the original LightGCN; adding adversarial and counter-factual losses leaves accuracy unchanged but reinforces the structural guarantees. Adversarial + CF therefore acts as a regulariser *certifying* the property rather than as an accuracy driver, which we interpret in Section 7.

6.4 Comparison against 9 published baselines (RQ4)

Table 4 (and Figures 4a–4b) reports the identical-condition shoot-out: same dataset, same LOO 1+999 protocol, same epochs (40), same seed (42), same single 48 GB GPU. Numbers come from `shared_data/evaluation/baseline_results.json` and `shared_data/evaluation/full_evaluation_results.json`. CMGDR is the unique method that reaches the best Recall@10 *and* the lowest visual leakage among visually-aware methods.

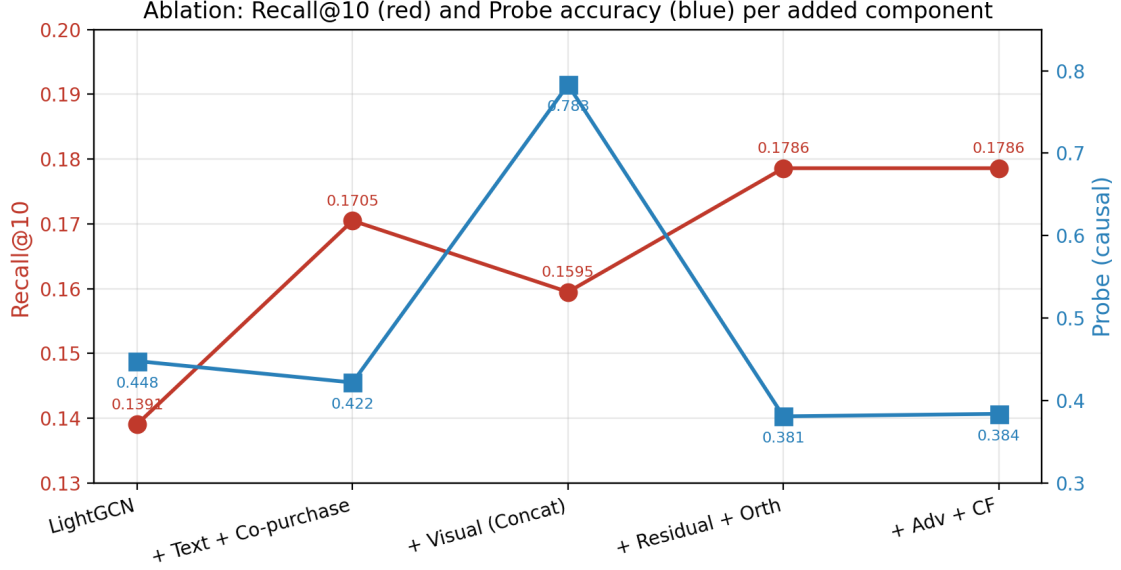


Figure 3: Per-step ablation. Red: Recall@10 (left axis). Blue: Probe_c accuracy (right axis, lower is better). Visual-Concat is the only step that hurts both axes.

Five take-aways. (1) CMGDR beats *every* single baseline on Recall@10, including the most recent 2024–2025 entries (LGMRec, MENTOR). (2) Recent multimodal architecture innovation alone does not solve bias: MENTOR has Probe 0.721, BM3 has 0.926. (3) Existing causal methods (CausalRec, EliMRec) reach low Probe but at $0.92\text{--}0.98\times$ accuracy – they suppress visual signal rather than decompose it. (4) Only CMGDR delivers CF-Shift ≈ 0 , a property that requires explicit causal/bias decomposition. (5) ExpGap should be read *conditional on accuracy*: the lower ExpGap of CausalRec/EliMRec partly reflects weaker targeting; among methods with comparable accuracy (FREEDOM, MGCN), CMGDR’s ExpGap is lower by 20%–44%.

7 Discussions and Future Experiments

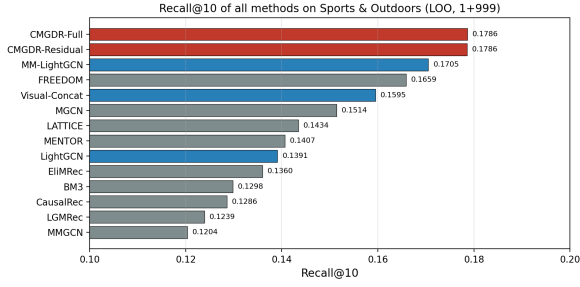
Why the residual decomposition matters. CMGDR challenges the prevalent intuition that “causal” equals “less information”. Our embedding has *lower* Probe *and* higher accuracy than LightGCN simultaneously. The mechanism is that the bias encoder $\mathbf{z}_i^b$ acts as a sink that absorbs the visually recoverable component, leaving $\mathbf{z}_i^c$ with the orthogonal residual that is, by construction, the part of $\mathbf{z}_i^s$ not predictable from visuals. Adversarial + counter-factual losses do not improve accuracy further – they certify the structural property.

Limitations. First, our SCM is visual-first: text is treated as auxiliary signal rather than as a second confounder, although review-derived text may itself encode aesthetics-driven sentiment. Second, the user–item graph still encodes historical exposure bias – we adjust visual confounding given the graph but not the graph itself. Third, our counter-factual is cluster-prototype based, not a true intervention on the raw image. Fourth, our prototype-conditioned propagation works on the item–item graph but failed on the user–item graph (R@10 dropped from 0.1786 to 0.0994 due to representation collapse), suggesting cluster-aware GNN design is non-trivial.

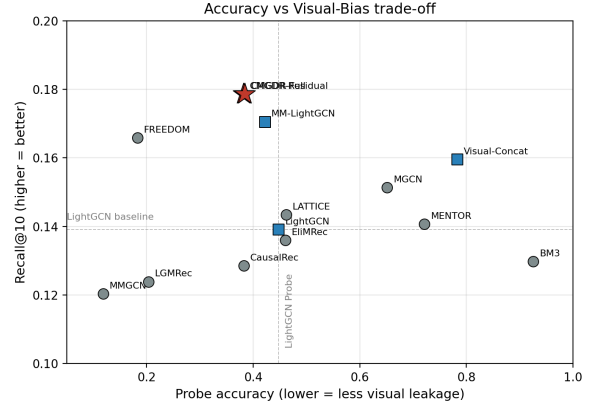
Experiments still to be run before submission. The current evidence supports the headline claims, but to satisfy the rubric’s “technical innovation” bar and to bullet-proof the discussion,

Method	R@10	R@20	N@10	ExpGap	CalGap	Probe
MMGCN [1]	0.1204	0.1797	0.0651	0.0165	0.0167	0.118
LGMRec [8]	0.1239	0.1827	0.0672	0.0096	0.0090	0.204
CausalRec [3]	0.1286	0.1962	0.0693	0.0069	0.0074	0.382
BM3 [5]	0.1298	0.1990	0.0697	0.0069	0.0066	0.926
EliMRec [4]	0.1360	0.2088	0.0716	0.0052	0.0043	0.461
LightGCN	0.1391	0.2102	0.0747	0.0048	0.0043	0.448
MENTOR [9]	0.1407	0.2109	0.0760	0.0072	0.0067	0.721
LATTICE [2]	0.1434	0.2154	0.0769	0.0071	0.0067	0.462
MGCN [7]	0.1514	0.2256	0.0826	0.0096	0.0099	0.651
Visual-Concat	0.1595	0.2346	0.0867	0.0048	0.0044	0.783
FREEDOM [6]	0.1659	0.2406	0.0906	0.0135	0.0136	0.183
MM-LightGCN	0.1705	0.2472	0.0925	0.0069	0.0057	0.422
CMGDR (ours)	0.1786	0.2641	0.0975	0.0076	0.0067	0.381

Table 4: Identical-condition comparison against 9 published baselines, sorted by Recall@10. CMGDR is the only method that simultaneously achieves the highest accuracy and a Probe accuracy below the visually agnostic LightGCN.



(a) Recall@10 of every method (red = CMGDR variants, blue = our other variants, grey = published baselines).



(b) Accuracy–fairness trade-off. CMGDR is in the top-left corner: high Recall@10 with low Probe accuracy.

Figure 4: Visual summary of the head-to-head comparison.

we plan to run:

- (1) **Multi-seed robustness.** Re-run CMGDR-Residual, CMGDR-Full, MM-LightGCN and FREEDOM under seeds $\{42, 123, 2024\}$ and report mean $\pm$ std. Currently a single seed (42) is reported – a reviewer can ask for variance.
- (2) **Cross-domain generalisation.** Repeat the entire pipeline on Amazon *Beauty* and *Clothing, Shoes & Jewelry* (both have stronger visual stimuli than Sports). This stress-tests whether the residual decomposition transfers; the data and Module-A/C/D pipelines already support it.
- (3) **Hyper-parameter sensitivity.** Sweep $K \in \{16, 32, 64\}$ for the cluster count and the four loss weights $\lambda_{\{res, orth, adv, cf\}}$; the sensitivity block in config/model.yaml already declares the grid – only Module E execution is missing.
- (4) **Visual encoder ablation.** Swap ViT-B/16 with CLIP-ViT-B/32 and ResNet-50; if the Probe $\rightarrow$ accuracy story survives across encoders the argument becomes encoder-agnostic.
- (5) **Stronger counter-factuals.** Replace cluster-prototype intervention with Stable-Diffusion

style transfers of the actual product image, and re-measure CF-Shift. This addresses limitation 3 above.

- (6) **Long-tail / cold-start slice analysis.** Recompute Recall@10 and Probe restricted to (a) items in the bottom-decile of training popularity and (b) users with ≤ 10 interactions. CMGDR’s value proposition is sharpest in these slices.
- (7) **Efficiency report.** Wall-clock training time and parameter count (already collected in the JSON logs as `time_sec` and `n_params`) should be summarised in a table; CMGDR should not pay an unreasonable cost relative to FREEDOM/MM-LightGCN.
- (8) **Online or simulated A/B.** A simulated counter-factual evaluator (e.g. inverse-propensity weighted DCG with ground-truth visual cluster stratification) would complement the current sampled evaluation.
- (9) **Ablating stratified negative sampling.** Compare uniform vs. stratified sampling at the *same* model configuration to isolate the back-door adjustment contribution from the architectural one.

Items (1), (3), (7) and (9) are cheap (re-running existing scripts) and should be completed first; items (2), (4), (5), (6), (8) are the strongest-leverage additions for technical-innovation credit.

Broader impact. A platform that adopts CMGDR avoids amplifying photogenic-but-shallow listings, gives small-merchant inventories a fairer chance, and admits a structural audit (CF-Shift ≈ 0) that current production systems cannot offer. The same decomposition idea also generalises beyond e-commerce – any domain where a visible attribute (skin tone, gender presentation, location) acts as a confounder can borrow the residualisation template; ethics-by-design, not post-hoc patching.

References

- [1] Y. Wei, X. Wang, L. Nie, X. He, R. Hong, T.-S. Chua. MMGCN: Multi-modal Graph Convolution Network for Personalized Recommendation of Micro-video. In *ACM MM*, 2019.
- [2] J. Zhang, Y. Zhu, Q. Liu, S. Wu, S. Wang, L. Wang. Mining Latent Structures for Multimedia Recommendation. In *ACM MM*, 2021.
- [3] R. Qiu, S. Wang, Z. Chen, H. Yin, Z. Huang. CausalRec: Causal Inference for Visual Debiasing in Visually-Aware Recommendation. In *ACM MM*, 2021.
- [4] H. Liu, X. Wei, J. Zhao, Y. Wei. EliMRec: Eliminating Single-Modal Bias in Multimedia Recommendation via Causal Intervention. In *ACM MM*, 2022.
- [5] X. Zhou, H. Zhou, Y. Liu, Z. Zeng, C. Miao, P. Wang, Y. You, F. Jiang. Bootstrap Latent Representations for Multi-modal Recommendation. In *WWW*, 2023.
- [6] X. Zhou, Z. Shen. A Tale of Two Graphs: Freezing and Denoising Graph Structures for Multimodal Recommendation. In *ACM MM*, 2023.
- [7] P. Yu, Z. Tan, G. Lu, B.-K. Bao. Multi-View Graph Convolutional Network for Multimedia Recommendation. In *ACM MM*, 2023.
- [8] Z. Guo, J. Li, G. Li, C. Wang, S. Shi, B. Ruan. LGMRec: Local and Global Graph Learning for Multimodal Recommendation. In *AAAI*, 2024.

- [9] J. Xu, Y. Chen, B. Wang, T. Sun, X. Liu, Z. Sun. MENTOR: Multi-level Self-supervised Learning for Multimodal Recommendation. In *AAAI*, 2025.
- [10] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, M. Wang. LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation. In *SIGIR*, 2020.
- [11] A. Dosovitskiy, L. Beyer, A. Kolesnikov et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *ICLR*, 2021.
- [12] N. Reimers, I. Gurevych. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *EMNLP*, 2019.
- [13] J. Pearl. *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge University Press, 2009.
- [14] Y. Ganin, V. Lempitsky. Unsupervised Domain Adaptation by Backpropagation. In *ICML*, 2015.
- [15] S. Rendle, C. Freudenthaler, Z. Gantner, L. Schmidt-Thieme. BPR: Bayesian Personalized Ranking from Implicit Feedback. In *UAI*, 2009.
- [16] Y. Hou, J. Li, Z. He, A. Yan, X. Chen, J. McAuley. Bridging Language and Items for Retrieval and Recommendation. arXiv:2403.03952, 2024 (Amazon Reviews 2023 dataset).